

Young Innovators · Week 1

555 Timer Alarm Siren Circuit

Build a flashing, beeping alarm with no code at all

Ages 12-14

No microcontroller

Falstad simulator



Did you know?

The 555 was designed in 1971. Billions have been made and it is still used today!





Look and listen

Blink, beep, blink, beep...



Car alarms

Flash and beep in a steady rhythm



Ambulance lights

Blink on and off, again and again



Kitchen timers

Beep when time is up



Many simple devices have no computer and no code. So what makes them keep perfect time?





Learning aims

By the end of this session, I can...



1



Explain what an **integrated circuit** is and identify the 555's 8 pins

2



Explain how a **capacitor** charging and discharging creates timing

3



Build a 555 **astable** alarm circuit on a breadboard

4



Measure voltages safely with a **multimeter**

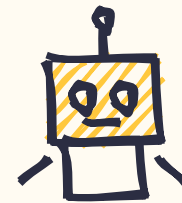
5



Calculate and change the **frequency** using R and C

Keywords

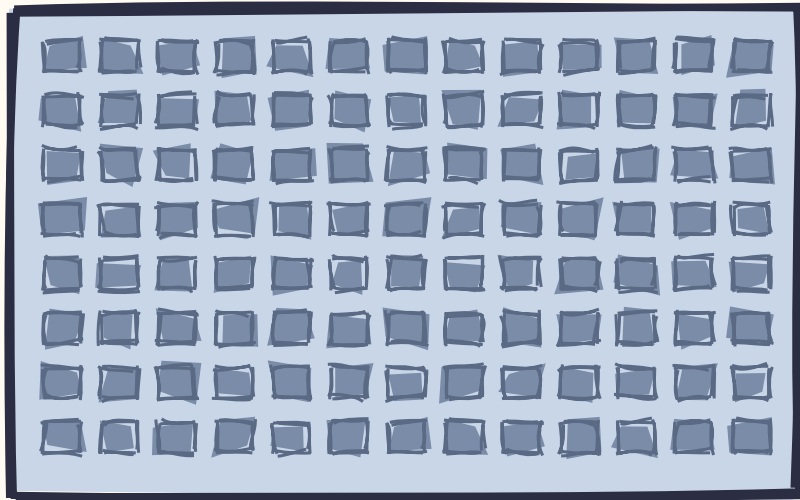
IC · capacitor · charge · astable ·
frequency · square wave · multimeter





What is a chip?

Integrated circuits (ICs)



tiny transistors and resistors on silicon

A whole circuit in one tiny package

An **integrated circuit** puts many transistors, resistors and connections onto one piece of **silicon**.

The 555 has about **25 transistors** inside. Your phone chip has **billions**.



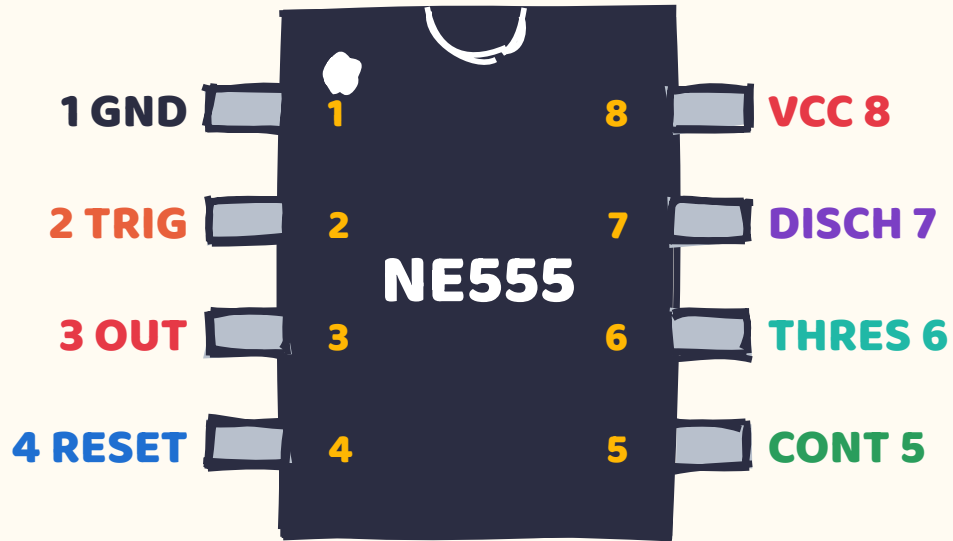
In real life

ICs are inside everything electronic: watches, cars, washing machines, traffic lights and toys.



Meet the star

The 555 timer: 8 pins, 8 jobs



Find pin 1

Look for the **notch** or **dot**. Pin 1 is next to it. Count **anti-clockwise** round the chip.

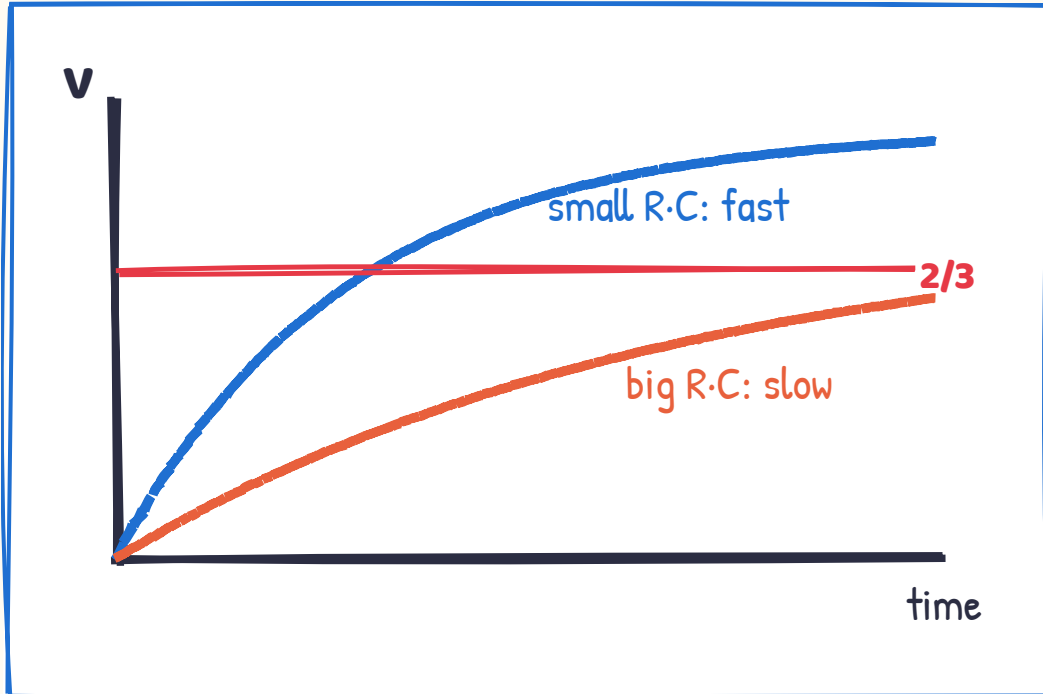
Pins we use today

- 8 power + · 1 ground
- 3 output (LED + buzzer)
- 2, 6, 7 timing (R and C)
- 4 reset (to +) · 5 10nF to ground



The timing secret

Capacitors: tiny charge buckets



Fill, then empty

A capacitor stores electric charge, like a bucket filling with water through a tap.

Bigger resistor = slower tap.

Bigger capacitor = bigger bucket.

Both make the timing **slower**.



In real life

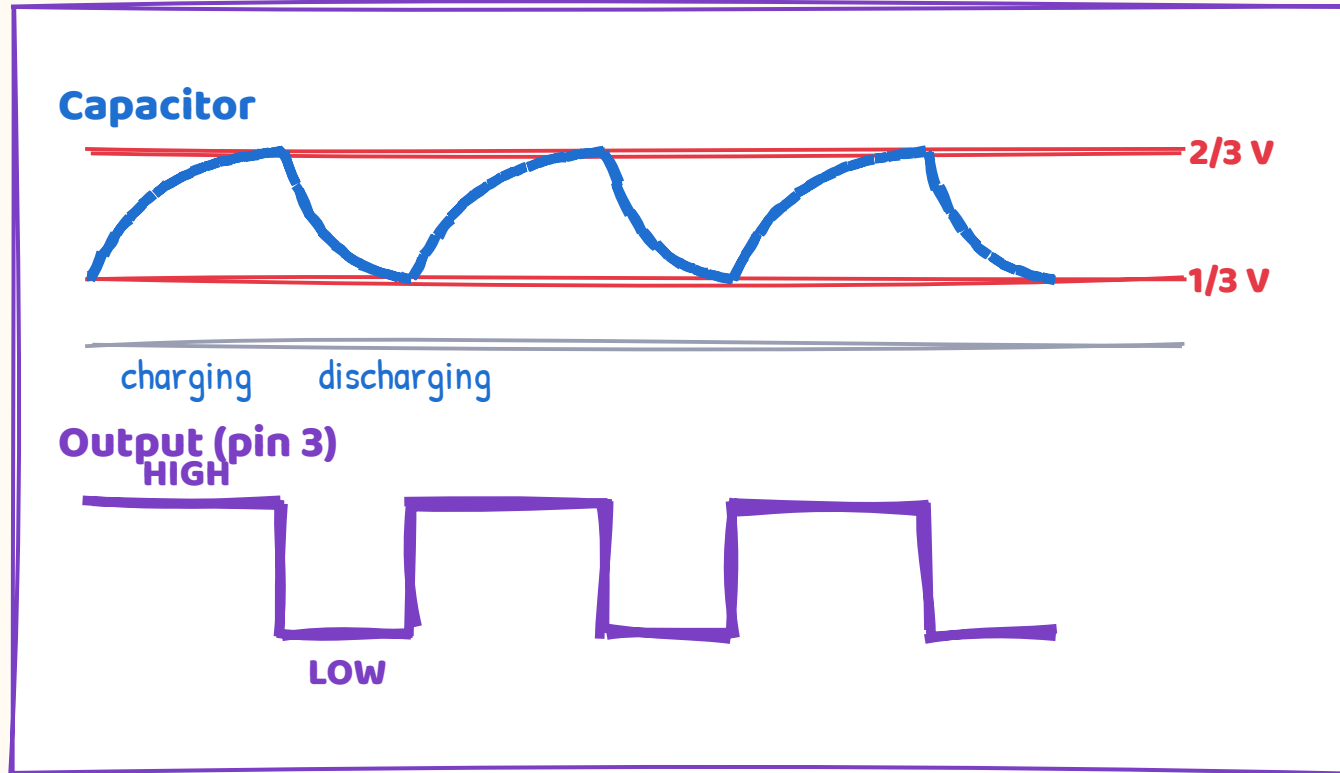
A camera flash charges a capacitor, then empties it in a split second for a bright flash.





How it works

Astable mode: a loop that never stops



1. The capacitor charges through R1 and R2. Output is **HIGH**: LED on, buzzer beeps.
2. At $2/3$ of the supply, the 555 flips. It discharges through R2. Output **LOW**.
3. At $1/3$, it flips back. Repeat forever!



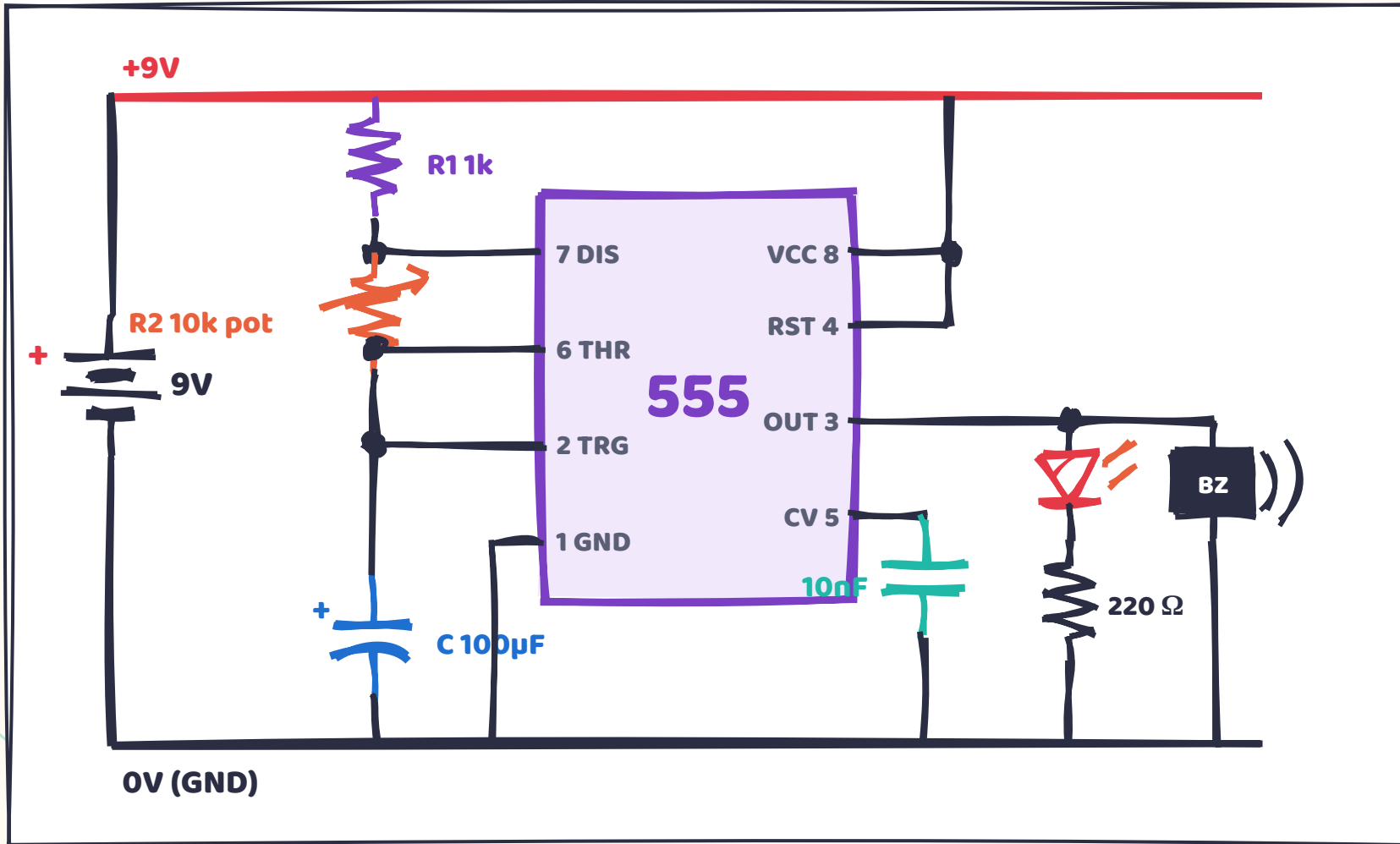
The output is a **square wave**: ON, OFF, ON, OFF... "astable" means it never settles.





The plan

Our alarm circuit (schematic)



Read the colours

Purple = timing resistor R1. Orange = speed knob R2. Blue = timing capacitor.

Teacher tip

Check the buzzer label: use one rated for 9V (many are 3-12V), or it may overheat.





Do the maths

How fast will it blink?



$$f = 1.44 / ((R1 + 2 \times R2) \times C)$$

f in hertz (Hz) = flashes per second · R in ohms · C in farads

Try it

Predict first, then test and time it with a stopwatch!

Worked example

R1 = 1,000 Ω, R2 = 10,000 Ω, C = 100 μF = 0.0001 F

$f = 1.44 / (21,000 \times 0.0001) = 1.44 / 2.1 =$

0.69 Hz

About one flash every **1.5 seconds**.

R1	R2	C	Frequency	You see / hear
1 k	10 k	100 μF	0.7 Hz	Slow flash
1 k	10 k	10 μF	6.9 Hz	Fast beeping
1 k	5 k	10 μF	13 Hz	Rapid buzz





Simulate first

Test it in the Falstad simulator

- 1 Open falstad.com/circuit and choose **Circuits** → **555 Timer Chip** → **Square Wave Generator**.
- 2 Change R2 and C to match our circuit.
- 3 Watch the capacitor voltage rise and fall on the scope.
- 4 Turn the speed up and down. Does your formula agree?



Why simulate?

Engineers test designs on a computer first. It is free, fast and nothing can burn!

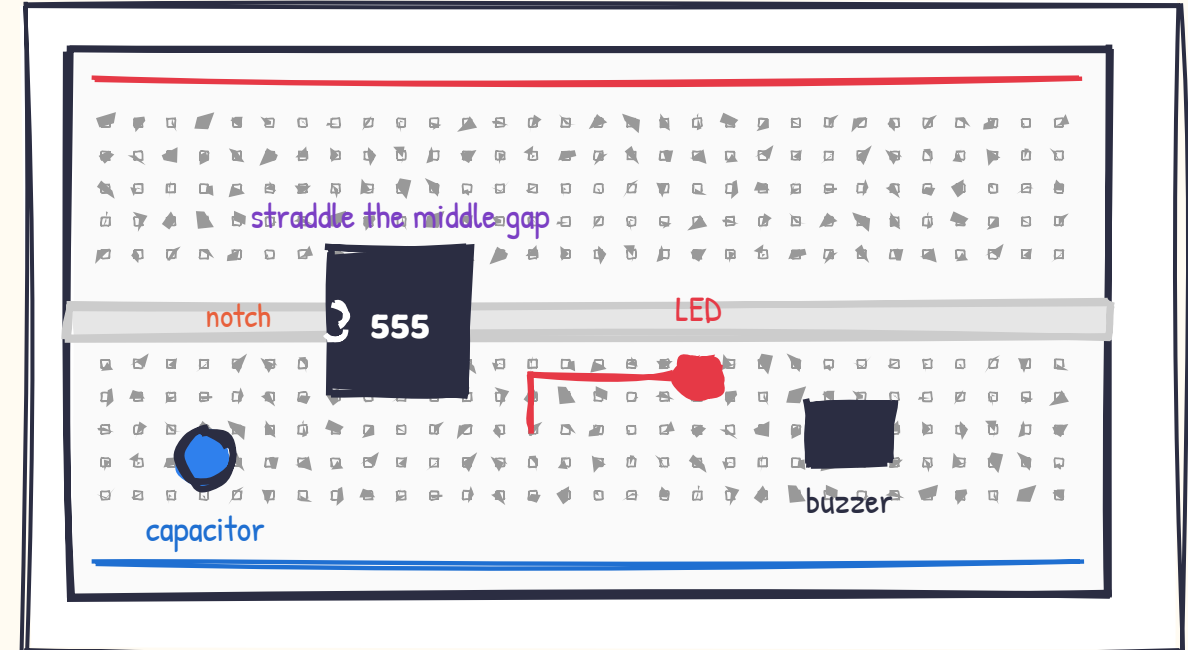


Build it

Breadboard steps



- 1 Place the 555 across the middle gap, notch on the left.
- 2 Pins 8 and 4 to the + rail; pin 1 to the - rail.
- 3 R1 from + to pin 7; R2 (pot) from pin 7 to pin 6.
- 4 Wire pin 6 to pin 2; capacitor from pin 2 to - (long leg to pin 2).
- 5 10nF from pin 5 to -.
- 6 Pin 3 to LED (with 220 Ω) and buzzer, then to -.
- 7 Teacher check, then connect the 9V battery.



Watch out

Electrolytic capacitors have a + and - side. The stripe marks the - leg.



Measure like an engineer

Multimeter checks



Measure	Expected	Your reading
Battery	about 9 V	_____
Pin 8 to pin 1	about 9 V	_____
Pin 3 (slow flash)	jumps high / low	_____
Capacitor (slow flash)	rises 3 V to 6 V	_____



How to measure

Dial to **20 V DC**. Black probe on - rail. Touch the red probe on each pin.

Use the **100 μ F** version so the numbers change slowly enough to read.



Think

Why does the capacitor stay between 3 V and 6 V on a 9 V supply? (Hint: 1/3 and 2/3!)





Discovery challenges

Make it your own

**Speed knob**

Turn R2. What happens to the flash rate? Explain why using the formula.

**Swap the bucket**

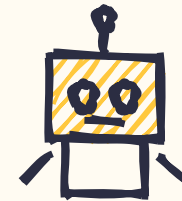
Change 100 μF to 10 μF . Predict the new frequency, then test.

**Police lights**

Add a second LED from + (with 220 Ω) to pin 3. The two LEDs now flash in turn!

**Your own alarm**

Design a rhythm for a bike alarm or a fire drill.

**Why it works**

When pin 3 is LOW, current flows from + through LED 2 into pin 3.





Test and debug

Bug hunter's guide

**Nothing happens**

Chip upside down? Check the notch. Is pin 4 connected to +?

**LED stays on**

Capacitor in the wrong way, or pin 2 and pin 6 not joined.

**Very dim LED**

Missing 220 Ω , or battery nearly flat: measure it.

**Chip gets hot**

Disconnect now! Usually a short from pin 3 or wrong power pins.

Engineer's habit
Unplug power before changing any wire.



Why it matters

The 555 in the real world



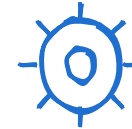
Car indicators

Older cars used timer circuits like this to blink



Toys and doorbells

Tones and melodies



Motor speed

Fast pulses control power (PWM)



Timers

Delays for lights and alarms



Big idea

Not every problem needs a microcontroller. A simple chip is **cheaper**, uses **less power** and **never crashes**. Good engineers choose the simplest tool that works.

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Industry & Innovation

Next week

YI-02: a heart rate monitor with a sensor and OLED screen.





Show what you know

Recap and exit ticket

Exit ticket

- ☒ Which pin is pin 1, and how do you find it?
- ☒ What two voltage levels make the 555 flip?
- ☒ What happens to the flash rate if C gets bigger?
- ☒ Calculate f for $R1 = 1k$, $R2 = 10k$, $C = 10 \mu F$.



Extension challenge

Use a second 555 to make a true two-tone siren: one slow timer switches a fast tone on and off.



Well done!

You built a timing circuit with no code, just physics!

